GELDIUM FINANCE

Exploratory Data Analysis (EDA)
Summary Report

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1. Introduction

This report reviews the structure, completeness, consistency, and early delinquency-related patterns in the supplied Geldium customer dataset before predictive modelling. The objective is to identify data quality concerns, recommend transparent treatment methods, and flag candidate risk indicators for validation in subsequent analysis. Findings below are descriptive and do not establish causation.

2. Dataset Overview

Attribute	Observed result
Dataset size	500 records and 19 columns
Customer identifiers	500 unique Customer_ID values; no duplicate full rows detected
Target variable	Delinquent_Account: 80 positive (1) and 420 negative (0); 16% positive class
Key variables	Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Employment_Status, Account_Tenure, Credit_Card_Type, Location, Month_1–Month_6
Data types	Numeric fields for financial/demographic measures; categorical fields for employment, card type, location, and six monthly payment-status fields

The Dataset Description Guide defines Month_1 to Month_6 as payment status (On-time, Late, Missed), Credit_Utilization as 0–100%, Credit_Score as typically 300–850, and Debt_to_Income_Ratio as a percentage.

Initial data-quality observations

- Missing values occur in Income (39/500, 7.8%), Loan_Balance (29/500, 5.8%), and Credit_Score (2/500, 0.4%). Other columns have no missing values.
- Employment_Status has inconsistent labels: 'Employed' (82), 'employed' (77), and 'EMP' (81) likely represent the same category; 'Self-employed' is also differently formatted from the other labels.
- Credit_Utilization is stored as a fraction for most records (e.g., median ≈ 0.486), although the guide describes it as a percentage. Four records exceed 1.0 (100%), reaching approximately 102.6%; confirm the unit and whether values above the credit limit are valid.
- Debt_to_Income_Ratio values range from about 0.10 to 0.553, suggesting a fraction/ratio representation rather than a 0–100 percentage representation. Confirm and document the unit before interpretation.
- No duplicate full rows were found, and Customer_ID is unique across all 500 records.

3. Missing Data Analysis

Variable / issue	Extent	Recommended handling	Reason
Income	39 missing (7.8%)	Investigate missingness; use median imputation as a baseline, preferably within training folds and with a missingness indicator if justified.	Income is skew-prone and median is robust to extremes. Assess whether missingness differs by target or customer segment before choosing the final method.
Loan_Balance	29 missing (5.8%)	Check source/system meaning; use median imputation as a baseline after checking missingness patterns.	Retains records while limiting sensitivity to extreme balances; do not impute blindly if missingness has business meaning.
Credit_Score	2 missing (0.4%)	Validate source records; if unrecoverable, use median imputation or exclude the two records in a sensitivity check.	Very small proportion is missing, but treatment should be compared and documented.
Employment_Status labels	Multiple variants for employment labels	Normalize case/whitespace and map confirmed synonyms (e.g., 'EMP' and 'employed') to a canonical category.	Prevents one real category being split into multiple misleading groups.
Credit_Utilization > 100%	4 records	Flag for source validation; do not cap or delete automatically.	May reflect over-limit balances or a unit/quality issue. Confirm the definition and business validity first.

No data was overwritten or imputed as part of this report. The methods above are recommendations for the data-preparation stage, pending source-owner confirmation and missingness analysis. Fit any imputation parameters on training data only to avoid leakage.

4. Key Findings and Risk Indicators

Observed patterns:

- The target is moderately imbalanced: 16% of records are labelled delinquent. Model evaluation should not rely on accuracy alone; recall, precision, F1, PR-AUC, and a confusion matrix should be considered.
- Numeric correlations with Delinquent_Account are very weak in this initial dataset: Income ≈ +0.045, Credit_Score ≈ +0.035, Debt_to_Income_Ratio ≈ +0.034, Credit_Utilization ≈ +0.034, Age ≈ +0.023, Loan_Balance ≈ −0.004, Missed_Payments ≈ −0.026, and Account_Tenure ≈ −0.040. These small correlations do not show strong linear associations in this sample.
- Unexpectedly, average Missed_Payments is slightly lower for the delinquent group (2.85) than for the non-delinquent group (2.99), while average Credit_Score is slightly higher for the delinquent group (about 591

versus 575). This is counterintuitive and should be investigated for target definition, data generation/quality, timing, or sampling issues before modelling.

- Monthly payment status categories are present in all six month columns, but their simple univariate association with the target is weak and inconsistent across months. Confirm chronological ordering and the target observation window before using these fields.
- The dataset guide lists age and location, but these variables should be handled carefully for fairness and governance. Their inclusion should be justified by legitimate use, applicable policy, and fairness assessment rather than assumed to be predictive or appropriate.

Candidate risk indicators to investigate further

Candidate variable	Why it may matter	Current evidence / caution
Payment history (Month_1–Month_6)	Recent late or missed payments may contain information about future repayment behaviour.	Validate timing and target window; observed patterns are not consistently aligned with the target in this dataset.
Missed_Payments	Prior missed payments may indicate repayment difficulty.	Current group averages run counter to the expected pattern; verify definitions and data quality.
Credit_Utilization	High usage of available credit may be associated with financial pressure.	Four values exceed 100%; unit convention needs confirmation.
Debt_to_Income_Ratio	Higher debt burden relative to income may reduce repayment capacity.	Values appear stored as fractions; confirm definition and scale.
Income and Loan_Balance	May help describe repayment capacity and outstanding obligations.	Both contain missing data; examine distributions and missingness before modelling.
Credit_Score and Account_Tenure	May summarize credit history and length of observed relationship.	Initial correlations with the target are weak; do not rank as strong predictors without further testing.

5. AI & GenAI Usage

GenAI is appropriate as an assistant for summarizing schema, drafting data-quality checks, proposing candidate imputation strategies, and helping document findings. All numerical findings in this report were checked against the supplied workbook. No synthetic records were generated and no missing values were filled in this report.

Example prompts used / suitable for this workflow:

- “Summarize the dataset structure, missing-value counts, data types, duplicate records, and suspicious values. Report counts and percentages, and distinguish observed findings from hypotheses.”
- “Suggest transparent handling methods for missing Income, Loan_Balance, and Credit_Score values. Explain assumptions, possible bias, and how to validate each method.”
- “Compare delinquency rates and numeric summary statistics across target classes. Flag counterintuitive relationships for investigation and do not infer causation.”

Confidentiality note: use only organization-approved AI tools for customer data. Avoid sharing direct identifiers or row-level sensitive financial information unless explicitly authorized. Aggregate or de-identify data where appropriate.

6. Conclusion & Next Steps

The dataset is structurally usable for further exploration (500 records, 19 columns, unique customer IDs, and no duplicate full rows), but it requires targeted data-quality review before predictive modelling. The main issues are missing Income, Loan_Balance, and Credit_Score values; inconsistent Employment_Status labels; and Credit_Utilization values above 100% alongside unclear ratio/percentage conventions. Most numeric associations with the target are weak, and some patterns (notably Missed_Payments and Credit_Score group averages) are counterintuitive. Recommended next steps are to confirm variable definitions and observation timing with Geldium, investigate missingness and anomalies, standardize confirmed category labels, document an imputation plan, and then evaluate candidate predictors using a leakage-safe train/test design and fairness checks.

Scope note: This is an initial EDA summary based on the supplied workbook and Dataset Description Guide. Risk indicators are candidates for validation, not causal conclusions or a deployed risk policy.